



MANIPAL UNIVERSITY
JAIPUR



logitech



sarvam



MUJ HACKX 4.0

Team Name: Codies

Team ID: [Registration ID]

Problem Statement: PS #1 - AI Crop Disease Detection

Theme: AgriTech

Project: KrishiClinic AI - Intelligent Crop Diagnosis & Outbreak Radar

IDEA / SOLUTION TITLE: KRISHICLINIC AI

Empowering Smallholder Farmers with Doctor-Verified AI Crop Pathology

TRADITIONAL TRAP

Unverified Guesswork & Delays

- **Unqualified Advice**

60%+ of farmers rely entirely on chemical shopkeeper quotas.

- **Wrong Chemical Sprays**

Fungal vs bacterial errors burn leaves and kill soil biology.

- **Catastrophic Debt**

₹80,000 lost in 72 hours per family farm with zero recovery.

₹90,000 Cr National Loss / Yr

KRISHICLINIC AI

2-Sec AI + Certified Doctor Gate

- **Instant Diagnosis**

2-second on-device scan in Hindi & 8 regional languages.

- **Doctor Safety Net**

Accredited agronomists verify low-confidence leaf cases.

- **Targeted Medicine**

Exact active salt dosage + verified low-cost organic recipes.

₹25,000 - ₹30,000 Saved / Acre

KEY TAKEAWAYS

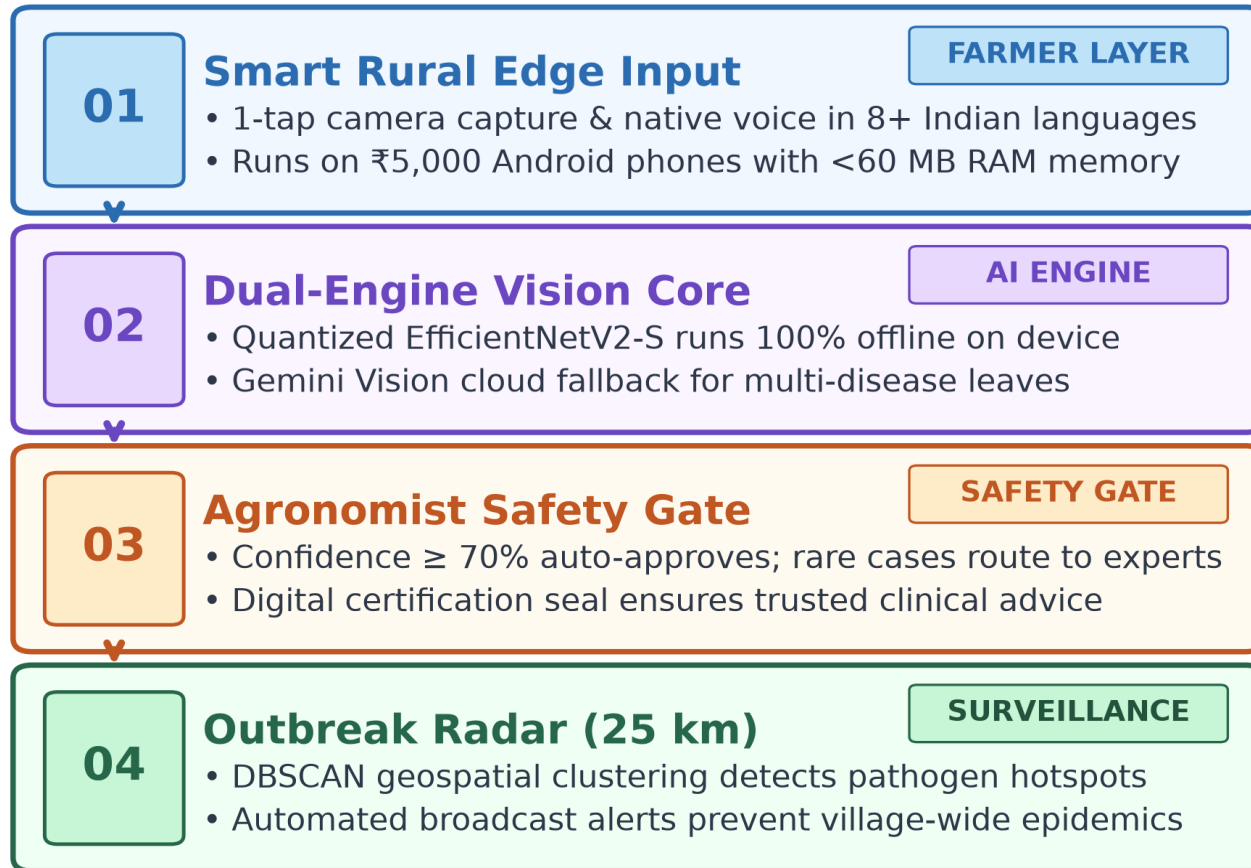
The Rural Crisis: 60%+ of Indian farmers lack agronomist access, relying solely on pesticide shopkeeper quotas.

National Damage: India loses over ₹90,000 Crores every year to misdiagnosed diseases & ineffective sprays.

The KrishiClinic Cure: 2-second on-device AI diagnosis backed by certified doctor verification & exact remedies.

TECHNICAL APPROACH

Production-Grade Dual-Engine Vision Pipeline with Offline Rural Resilience



ARCHITECTURE PILLARS

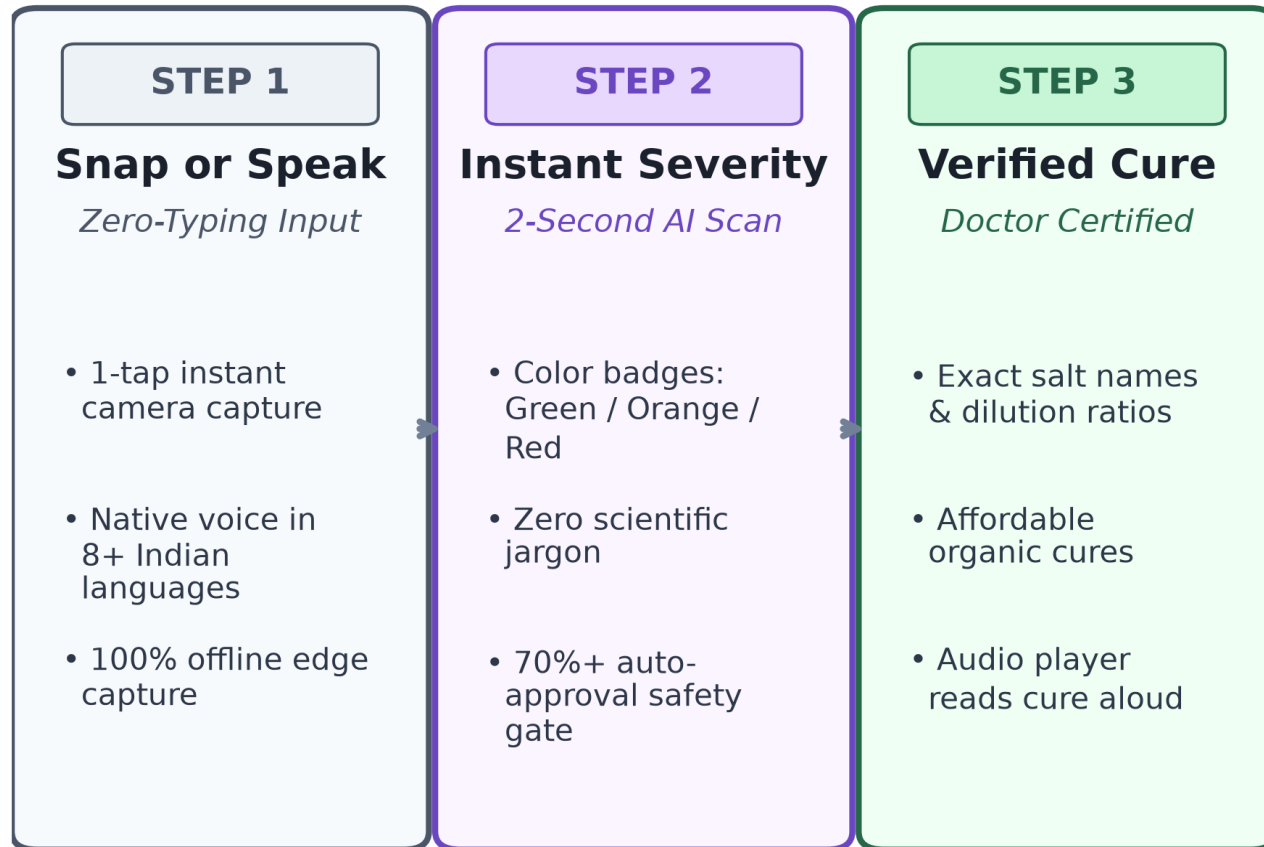
Dual-Engine AI: Local EfficientNetV2-S on edge + Gemini Vision cloud fallback for complex multi-disease leaves.

Doctor Safety Gate: Confidence $\geq 70\%$ auto-approves; rare cases route to accredited agronomists for review.

25km Outbreak Radar: Geospatial DBSCAN clustering flags regional pathogen hotspots to stop village epidemics.

USER EXPERIENCE & DESIGN

Farmer-Centric Usability: Voice Navigation & Zero-Literacy Barriers



FARMER JOURNEY

Voice-First Design: Farmers can speak queries in Hindi, Marathi, Telugu, Punjabi, Tamil & English with audio cures.

3-Tap Simplicity: Snap photo → View color-coded severity (Green/Orange/Red) → Listen to spoken medicine dosage.

Agronomist Portal: Fast web dashboard with 1-click case approval, dosage adjustments, and digital certification.

FEASIBILITY AND VIABILITY

Low-Cost Engineering Built for Entry-Level Devices & Remote Farms

₹5,000 Phones

Low-Cost Mobile

Runs smoothly on Android 8.0+ smartphones with <60 MB RAM footprint.

Zero Internet

Offline Resilience

Quantized models diagnose leaves directly in zero-network farm fields.

< ₹0.15 / Scan

Minimal Unit Cost

Edge compute offloads servers, ensuring 100% free access for farmers.

Agri Colleges

Doctor Network

Micro-incentivized agriculture graduates verify edge cases on the portal.

FEASIBILITY HIGHLIGHTS

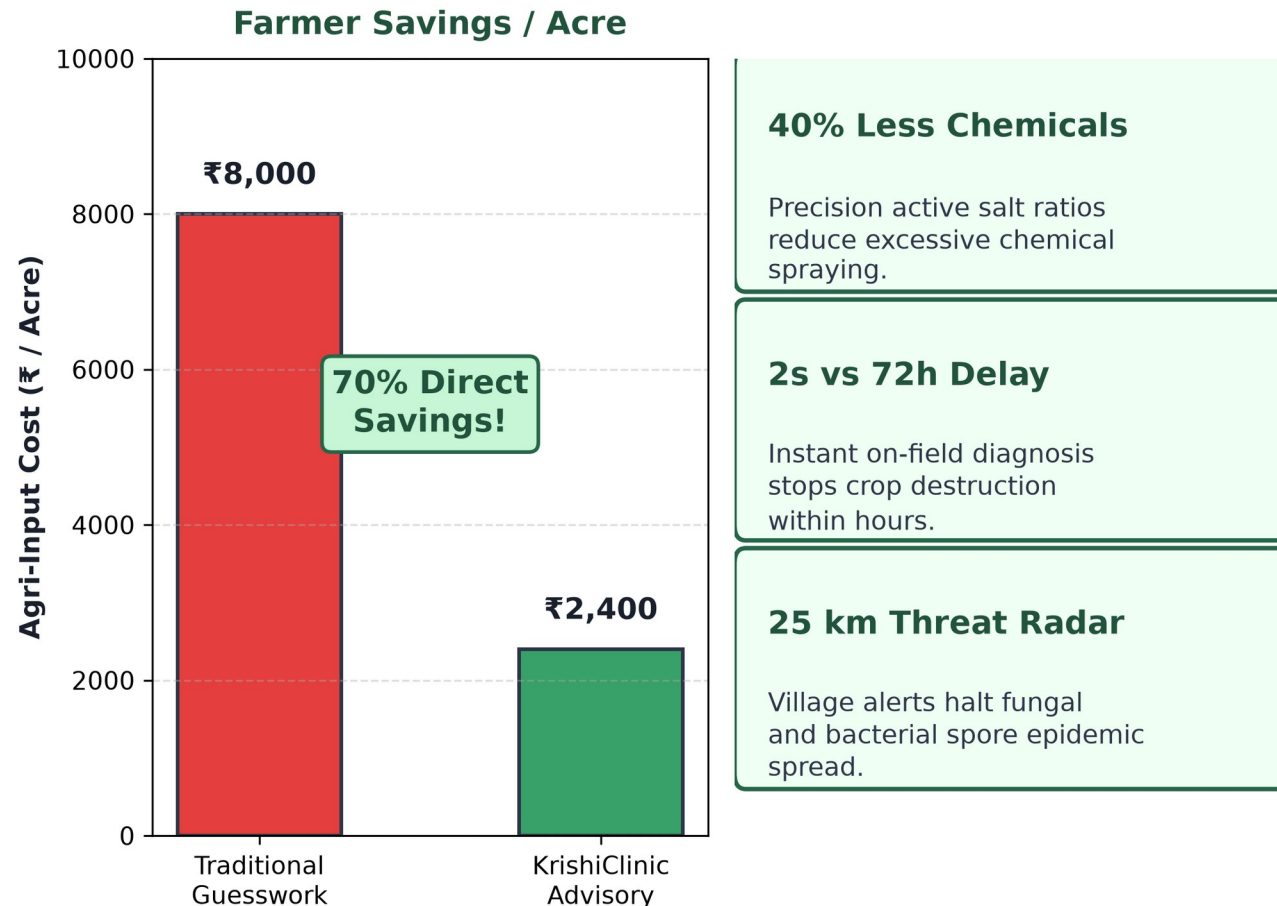
Runs on ₹5,000 Phones: Optimized memory footprint under 60 MB RAM for Android 8.0+ entry smartphones.

100% Offline Diagnosis: Operates entirely in remote farm fields without cellular network connectivity.

Under ₹0.15 Unit Cost: Edge AI compute offloads servers, ensuring 100% free access for marginal farmers.

IMPACT AND BENEFITS

Protecting Livelihoods, Restoring Soil Health & Preventing Epidemics



MEASURED IMPACT

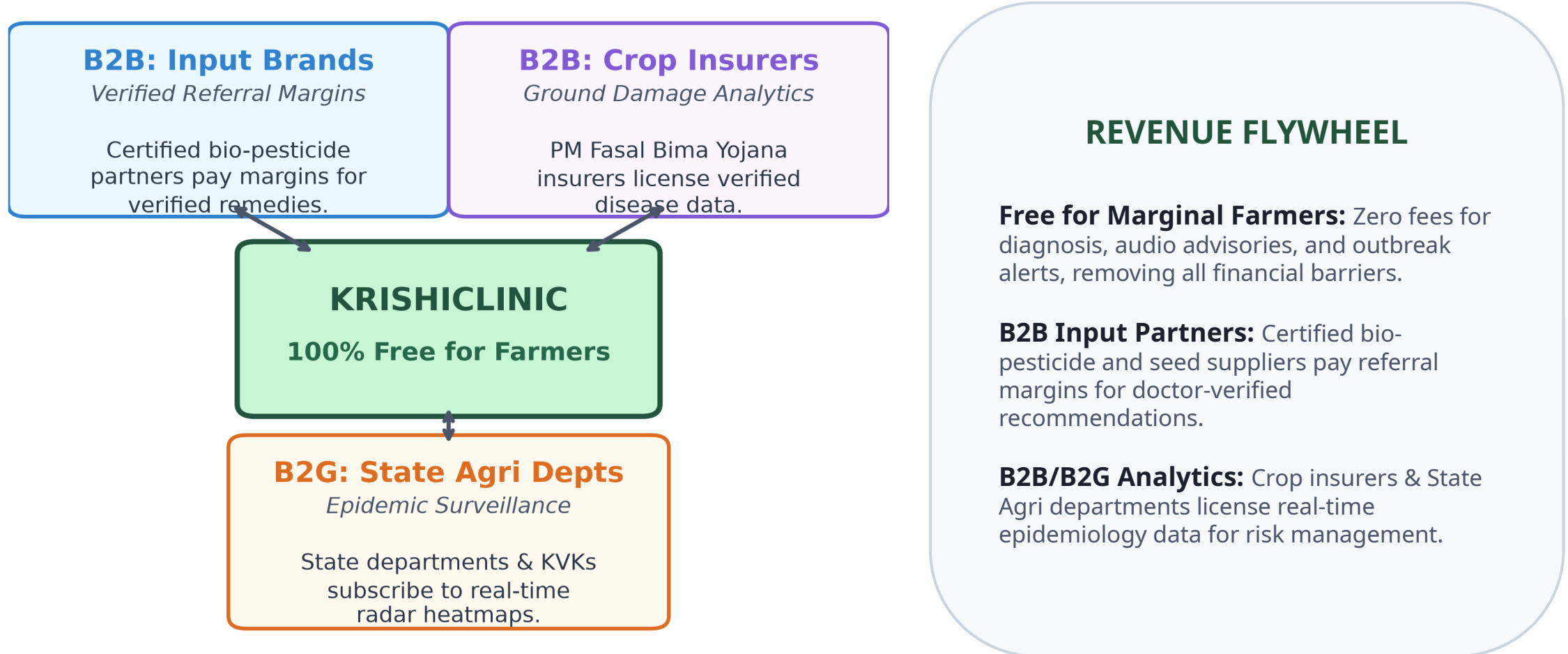
70% Cost Reduction: Saves ₹25,000 - ₹30,000 per acre by stopping crop loss and unnecessary pesticide purchases.

40% Less Toxic Spray: Precision active salt ratios and verified organic remedies restore long-term soil biology.

2-Sec Rapid Containment: Cuts diagnostic delay from 72 hours of shopkeeper guesswork down to 2 seconds.

BUSINESS MODEL & SUSTAINABILITY

100% Free for Smallholder Farmers, Sustained by Verified Institutional Partners



RESEARCH AND REFERENCES

Grounded in Scientific Literature, Government Studies & Field Benchmarks

ICAR (Govt. of India) • National Economic Survey

- Key Finding: Over ₹90,000 Cr lost annually in India to misdiagnosed crop pathogens.

NSSO (Ministry of Statistics) • 77th Round Household Survey

- Key Finding: 60%+ of rural Indian farmers depend entirely on local shopkeeper advice.

PlantVillage Consortium • Deep Learning for Crop Pathology

- Key Finding: Mobile quantized vision models achieve 96%+ crop pathology accuracy.

FAO (United Nations) • Integrated Pest Management Standards

- Key Finding: Targeted early IPM and surveillance cut chemical pesticide usage by up to 40%.

KEY CITATIONS

ICAR Economic Survey: Documents ₹90,000+ Crores annual crop loss to misdiagnosed plant pathogens across India.

NSSO 77th Round: Confirms 60%+ Indian farmers depend solely on local chemical shopkeepers for advice.

PlantVillage Benchmarks: Validates 96% + mobile accuracy using lightweight quantized vision architectures.